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# Where Does Platonic Convergence Live? A Linear Readout Gains Where the Galaxy Embedding Bends Toward the Label and Loses Where It Bends Away

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## Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by  
2 measuring the convergence of foundation-model representations of galaxy images  
3 with mutual  $k$ -nearest-neighbour agreement (MKNN); we ask whether that con-  
4 vergence concentrates where the embedding manifold is flat. A pre-registered  
5 raw-decoder mean-curvature statistic is negatively associated with cross-survey  
6 agreement, but density removes most of the signal and the one residual case dis-  
7 appears once its radial component is removed. A classical identity, expanded to  
8 second order on the manifold, explains why: a readout sees  $\langle w_N, \Pi \rangle$ , the second  
9 fundamental form contracted with its normal component; under first-order tangent  
10 matching an alignment’s second-order error decomposes into a connection mis-  
11 match and a relative extrinsic term; mean curvature is only the metric trace of  
12  $\Pi$  and does not in general determine either. On 86,471 labelled galaxies a linear  
13 readout gains accuracy where the manifold bends toward the label’s own curvature  
14 and loses it where the manifold bends away. In a counterfactual local second-order  
15 surrogate of the readout, the manifold’s bending in the readout’s normal direction  
16 raises local  $R^2$  at 79–99% of anchors, its sign reversal lowers it at 97–100%, and  
17 a random orientation of matched strength shows no such asymmetry and is typi-  
18 cally harmful under either sign, on five encoders; across anchors, the mismatch  
19 between the label Hessian and the probe-facing Hessian is negative wherever the  
20 label has local structure and survives cross-fitting and a change of encoder. Mean  
21 curvature of a single representation does not explain weak Platonic convergence;  
22 the obstruction is relative and, for the fitted global linear alignment, present at first  
23 order.

## 24 1 Introduction

25 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data  
26 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on galaxies  
27 imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC), which share only 0.3 to 2.3  
28 percent of nearest neighbours across surveys under a family of vision models. If two models disagree  
29 most where the embedding surface bends sharply, local mean curvature should associate negatively  
30 with local cross-survey agreement.

31 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a  
32 trained decoder, validated against analytic truth; (ii) a density-controlled test of the curvature–  
33 alignment prediction, which finds no density-free effect; and (iii) the classical identity  $\text{Hess}_M(wx) =$   
34  $\langle w_N, \Pi \rangle$  applied to a fitted probe on learned manifolds: the estimated pointwise tensor, compared  
35 with the label’s own Hessian, predicts local readout accuracy on five encoders, and an intervention

on that tensor with a matched control tests the mechanism, so a readout gains where the manifold bends toward the label’s curvature and loses where it bends away.

## 2 Pointwise curvature from a decoder

**Data.** DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019] (Platonic Universe release [Duraphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey galaxies embedded by a supervised ImageNet-21k ViT-B/16 (“ViT-B” below; not DINOv3) with catalogue labels ( $r$ -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment. We unit-normalize embeddings (data on  $S^{767}$ ).

**Instrument.** Let  $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$  be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; we evaluate curvature on held-out rows. With  $J = DF(z)$ , metric  $g = J^\top J$ , normal projector  $P_N = I - Jg^{-1}J^\top$  and second fundamental form  $\Pi = P_N D^2 F(z)$ , the mean-curvature vector of the raw decoder surface is  $H_{\text{raw}} = \text{tr}_g \Pi$  (un-normalized trace; some authors divide by  $d$ ), by automatic differentiation (cf. [Lee and Park, 2023, Acosta et al., 2023]), at points where  $DF$  has full column rank  $d$  (likewise  $D(F/\|F\|)$  after projection). On the unit sphere the Euclidean mean-curvature vector of every  $d$ -submanifold has the universal radial component  $-d \hat{x}$ , which says nothing about shape; differentiating  $F/\|F\|$  instead splits  $H$  exactly as  $-d \hat{x} + H^S$  ( $H_{\text{rad}} = -d$  to  $10^{-14}$ ), and  $H^S$  carries the non-radial mean-curvature component. Section 3 uses the pre-registered  $\|H_{\text{raw}}\|$ ; Sections 4–5 use the projected decoder.

**Validation.** Against closed-form curvature, at  $D = 768$  an in-sphere surface  $G(z) = Q \text{ normalize}([\phi(z); 0.8 h(z); 0 \dots])$  ( $\phi$  inverse stereographic  $\mathbb{R}^{16} \rightarrow S^{16}$ ,  $h$  four Gaussian bumps,  $Q$  a rotation) gives rank 0.94 against the true  $\|H^S\|$ , direction cosine 0.999 and magnitude ratio 1.00 for the full vector.

## 3 Curvature and cross-survey convergence

We correlate the per-point mutual  $k$ -nearest-neighbour score  $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$  at  $k = 20$  between the HSC and Legacy Survey embeddings with  $\|H_{\text{raw}}\|$  at  $d \in \{20, 25, 32\}$ . The raw association  $\rho(\|H_{\text{raw}}\|, \text{MKNN})$  is  $-0.112$ ,  $-0.128$  and  $-0.024$  at  $d = 20, 25, 32$  ( $p < 10^{-5}$ ,  $< 10^{-5}$ ,  $0.009$ ). But  $k$ -NN statistics are density-sensitive [Radovanović et al., 2010]: local density ( $k = 30$  estimate on the Legacy embedding) correlates with agreement at  $-0.21$  at every  $d$  and with  $\|H_{\text{raw}}\|$  at  $+0.43$ ,  $+0.32$  and  $+0.01$ , so residualizing on density leaves  $-0.024$ ,  $-0.066$  and  $-0.022$ . Under a permutation null within density strata only  $d = 25$  survives ( $p < 5 \times 10^{-5}$ ), and subtracting the radial projection of  $H_{\text{raw}}$  (a projection of  $H_{\text{raw}}$ , not a recomputation on  $F/\|F\|$ ) collapses it to  $-0.023$ .

## 4 What a linear map sees on a curved manifold

Let  $M \subset \mathbb{R}^D$  be a  $d$ -dimensional submanifold,  $d < D$ ; at  $x_0$  take an orthonormal tangent frame  $J$ , the second fundamental form  $\Pi$  (normal-valued, symmetric  $d \times d$ ),  $H = \text{tr } \Pi$ , and for a vector  $w$  its tangent and normal parts  $w_T + w_N$ . In normal coordinates  $u$  at  $x_0$ ,  $M$  is  $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$  [Lee, 2018], so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

For  $M \subset S^{D-1}$ ,  $\Pi = \Pi^S - g \hat{x}$  splits off the sphere’s radial direction, so  $\text{Hess}_M(w \cdot x) = \langle w_S, \Pi^S \rangle - (w \cdot \hat{x}) g$  (2), with  $w_S$  the part of  $w_N$  orthogonal to  $\hat{x}$ : the sphere bends every readout of a unit-normalized embedding in proportion to its own value  $\hat{y} - b_0$ . For a label  $y$  on  $M$ , a global affine probe  $\hat{y} = w \cdot x + b_0$  and a patch  $u \sim N(0, s^2 I_d)$  in normal coordinates, embedding and label locally  $C^4$ , with  $\bar{c}$  the patch-mean residual,  $\delta = [\nabla_M y]_J - J^\top w$ ,  $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$  (all tensor norms and cosines use  $\langle A, B \rangle_g = g^{ia} g^{jb} A_{ij} B_{ab}$ ),

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_g^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

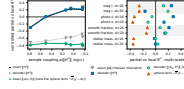


Figure 1: (a) Known surface, linear label, exact geometry: three-control partial vs local  $R^2$  as the density-curvature coupling is set by construction ( $\gamma$  from  $-1$  to  $1$ ). (b) Galaxies, 512 anchors,  $k = 2,048$ , multi-scale control: the decoder's  $\|H^S\|$  (filled squares), shape term (open circles), sphere term (open triangles).

| label           | $\ H^S\ $ |        | shape $\ \langle w_S, \Pi^S \rangle\ $ |        | sphere $\sqrt{\hat{d}} \hat{y} - b_0 $ |       | mismatch $\ \Delta\ _g$ |        | alignment |        |
|-----------------|-----------|--------|----------------------------------------|--------|----------------------------------------|-------|-------------------------|--------|-----------|--------|
|                 | $d=16$    | $d=20$ | 16                                     | 20     | 16                                     | 20    | 16                      | 20     | 16        | 20     |
| mag_r           | +0.25     | -0.17  | +0.12                                  | +0.19  | -0.26                                  | -0.26 | -0.39                   | -0.40  | +0.35     | +0.43  |
| photo_z         | +0.28     | +0.13  | -0.06*                                 | -0.12  | -0.33                                  | -0.36 | -0.45                   | -0.44  | +0.26     | +0.24  |
| smooth_fraction | +0.21     | +0.15  | +0.07*                                 | +0.16  | -0.13                                  | -0.15 | -0.09*                  | -0.17  | +0.19     | +0.07* |
| stellar_mass    | -0.02*    | -0.00* | +0.00*                                 | +0.03* | -0.35                                  | -0.35 | -0.04*                  | -0.09* | +0.01*    | +0.02* |

Table 1: Galaxies, 512 anchors,  $k = 2,048$ : rank-partial Spearman against local  $R^2$  under the multi-scale density control. \* not significant at 0.05.

For two embeddings  $F, G$  of the same objects in a common chart  $u$  and a linear map  $A$  with  $AJ_F = J_G$ , the same expansion gives  $G - (G_0 + A(F - F_0)) = \frac{1}{2}[J_G(\Gamma^G - \Gamma^F) + \Pi_G - A\Pi_F](u, u) + O(|u|^3)$ , with  $\Gamma$  the induced connections: the second-order residual contains a connection mismatch and a *relative* extrinsic term, whose target-normal component is  $\Pi_G - P_N^G A \Pi_F$ .

So  $\|H\|$  need not correlate with either term; and since the second-order term scales as  $s^4$  with  $s$  set by density at fixed  $k$ , patch-scored curvature effects are density-sensitive.

## 5 Probe-facing curvature: bending toward the label helps, bending away hurts

Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered  $\alpha = 100$  from embedding to label; local  $R^2$  inside each of 512 anchors' 2,048-neighbourhoods; rank-partial Spearman against local  $R^2$  with Freedman-Lane  $p$ -values [Freedman and Lane, 1983], controlling for log patch radius, local label variance and count and, stronger, log radius at five scales.

**Known surface.** Resampling the validation surface with weight  $\|H^S\|^\gamma$  (Figure 1a) moves the  $\|H^S\|$  partial from  $-0.29$  to  $+0.26$  while  $\|\langle w_N, \Pi \rangle\|_g$  holds at  $-0.74$  to  $-0.80$ , dominated by the sphere term of Eq. (2).

**Galaxies.** Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99). We estimate  $\text{Hess}_M y$  per anchor by a least-squares quadratic of the label on the tangent-projected coordinates  $u = g^{-1}J^\top(x - x_0)$  of its 2,048 neighbours (decoder-based Monge coordinates, so the finite-patch coefficient estimates the covariant Hessian). Three regularities follow (Table 1; Appendix A); the sphere term is negative on every label and both  $d$  ( $-0.13$  to  $-0.36$ ) under the sealed probe; with a weak-ridge probe ( $\alpha = 1$ , Appendix B) it stays negative for magnitude and smooth fraction ( $-0.21$ ,  $-0.12$ ) and vanishes for redshift and stellar mass. For magnitude, redshift and smooth fraction the mismatch is negative ( $-0.09$  to  $-0.45$ ; unchanged under cross-fitting, Appendix B) and the alignment  $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$  positive ( $+0.07$  to  $+0.43$ ); stellar mass shows no mismatch or alignment signal here. The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift): with  $K = \langle w_N, \Pi \rangle$ ,  $\|\Delta\|_g^2 = \|\text{Hess}_M y\|_g^2 - 2\langle \text{Hess}_M y, K \rangle_g + \|K\|_g^2$ , so the cross term decides for infinitesimal scaling from zero and  $\|K\|_g^2$  is the finite-magnitude overshoot penalty. The same regression on the readout  $w \cdot x$  returns the point cloud's own  $\langle w_N, \Pi \rangle$  (median cosine with the decoder's 0.55–0.65).

**Bending toward helps, bending away hurts.** Within the quadratic surrogate of Eq. (3), the Hessian-mismatch term is lower than at  $K = 0$  iff  $2\langle \text{Hess}_M y, K \rangle_g > \|K\|_g^2$ : the manifold must bend toward the label's own curvature, and not too far. (The identity is classical, and Liu et al. [2023]

already derive, for least squares on a hypersurface, a normal weight proportional to the label’s second derivatives over curvature, the low-dimensional special case of this mechanism; curved representations distorting a linear decoder is documented [Jurewicz et al., 2024],  $\text{Hess}_M y$  is a classical object [Donoho and Grimes, 2003], and manifold capacity [Chung et al., 2016, Slatton et al., 2026], flattening [Psenka et al., 2024], curvature regularization [Ma et al., 2023], latent-curvature studies [Kaufman and Azencot, 2023], local-regression bias [Cheng and Wu, 2013] and geometry markers of generalization [Chou et al., 2026] use dimension, radius or magnitude. New here is the estimated pointwise tensor  $K$  on a learned manifold, its tensor comparison with  $\text{Hess}_M y$ , and the intervention that follows.) No anchor is flat, so we test the condition with a counterfactual local second-order surrogate of the readout. Writing  $K = K_S - (w \cdot \hat{x})g$  with  $K_S = \langle w_S, \Pi^S \rangle$  the in-sphere shape term, each anchor’s 2,048 neighbours are scored by  $\hat{y}_t(u) = c + a^\top u + \frac{1}{2} K_{\text{sph}}(u, u) + \frac{t}{2} K_S(u, u)$  with  $a$  the readout’s tangent gradient,  $K_S$  from the decoder and  $c$  refit:  $t = 1$  is the manifold’s bending as seen in the readout’s normal direction (“bending toward the label” means  $\langle R, K_S \rangle > 0$  for the remaining label curvature  $R = \text{Hess}_M y - K_{\text{sph}}$ ),  $t = -1$  the sign reversal of that probe-facing shape quadratic,  $t = 0$  shape-flat. In the Gaussian model  $t = 1$  beats shape-flat iff  $2\langle R, K_S \rangle_g > \|K_S\|_g^2$ ; on the actual neighbours, with  $r_0$  the centred shape-flat residual and  $q_c$  the centred quadratic  $\frac{1}{2} K_S(u_i, u_i)$ , the exact criterion is  $2\langle r_0, q_c \rangle > \|q_c\|^2$  with optimum  $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$  (Appendix E). On five encoders and all four labels,  $t = 1$  beats shape-flat at 79–99% of anchors and  $t = -1$  is worse at 97–100%, while a random orientation with the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign ( $t = 1$  beats shape-flat at 22–36% of anchors,  $t^* \approx 0$ ; Appendix E). The probe-selected shape quadratic is locally error-reducing; under the second-order model that is the signature of bending toward the remaining label curvature. This within-anchor question differs from Table 1’s cross-anchor one (does variation in mismatch explain variation in local  $R^2$ ) and can hold where that has no signal, as for stellar mass.  $t^*$  sits at 1.5 to 3.6 throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial second-order term.

**Across encoders.** Across five encoders (the readout set’s supervised ViT-B, DINOv3 ViT-B/16, CLIP-B, ConvNeXt-B, ViT-L;  $d = 16$ ) the quadratic-chart mean-bending trace has density-controlled associations with the magnitude probe’s local  $R^2$  of both signs (−0.24 to +0.27), as Eq. (1) allows. Per encoder (Appendix D) the decoder test is stable where the expansion predicts: the mismatch partial is negative for magnitude and redshift on all five encoders (−0.32 to −0.61; cross-fitted −0.28 to −0.62) and the alignment positive for magnitude on all five (+0.13 to +0.53) and for redshift on four; the shape term’s sign varies with encoder as well as label; for the mismatch and alignment hypotheses stellar mass stays null but for one mismatch cell (−0.12).

## 6 Discussion

A linear readout of a galaxy embedding gains accuracy where the manifold, seen in the readout’s normal direction, bends toward the label’s curvature and loses it where that bending is reversed: gain at a large majority of anchors and loss at nearly every anchor, on five encoders. The residual expansion names the quantity, and a decoder’s pointwise second fundamental form measures it at production scale, validated against analytic truth for  $H^S$  and in substantial agreement with the data. Under first-order tangent matching, the relevant second-order extrinsic normal term of a linear alignment map [Bangachev et al., 2026] is  $\Pi_G - P_N^G \Pi_F$ , not one embedding’s mean curvature; the full residual also carries the connection mismatch; our pilot (Appendix C) finds a 44% first-order tangent-frame residual, which leaves its relative-II column only an extrinsic diagnostic.

**Limitations.** One survey pair. The label’s Hessian is a local estimate; its norm is reliably ranked (split-half  $\rho$  0.91–0.94) but its direction only moderately (tensor cosine 0.19–0.35). On a synthetic surface with known curvature the mismatch partial reproduces but the alignment partial has no stable sign; Appendix E carries the alignment claim. Mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation  $p$ -values assume anchor exchangeability under overlapping patches.

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## 223 Appendix A: decoder ablations

224 Table 2 repeats the four quantities of Table 1 at  $d = 16$  under the multi-scale density control for two  
225 further decoder initialisation seeds and for a decoder of width  $400^3$  instead of  $250^3$  (all fits reach  
variance explained 0.952). Every significant sign of the main table is retained.

| label           | quantity  | seed 0 | seed 1 | seed 2 | $400^3$ |
|-----------------|-----------|--------|--------|--------|---------|
| mag_r           | shape     | +0.12  | +0.05* | +0.12  | +0.05*  |
|                 | sphere    | −0.26  | −0.26  | −0.27  | −0.26   |
|                 | mismatch  | −0.39  | −0.35  | −0.34  | −0.34   |
|                 | alignment | +0.35  | +0.31  | +0.37  | +0.41   |
| photo_z         | shape     | −0.06* | −0.09* | −0.09  | −0.06*  |
|                 | sphere    | −0.33  | −0.35  | −0.34  | −0.35   |
|                 | mismatch  | −0.45  | −0.43  | −0.42  | −0.41   |
|                 | alignment | +0.26  | +0.24  | +0.25  | +0.25   |
| smooth_fraction | shape     | +0.07* | +0.20  | +0.15  | +0.10   |
|                 | sphere    | −0.13  | −0.15  | −0.14  | −0.11   |
|                 | mismatch  | −0.09* | −0.05* | −0.08* | −0.07*  |
|                 | alignment | +0.19  | +0.14  | +0.09  | +0.10   |
| stellar_mass    | shape     | +0.00* | −0.06* | +0.05* | +0.09   |
|                 | sphere    | −0.35  | −0.35  | −0.35  | −0.35   |
|                 | mismatch  | −0.04* | +0.03* | +0.02* | +0.03*  |
|                 | alignment | +0.01* | −0.02* | +0.03* | −0.02*  |

Table 2: Decoder ablations,  $d = 16$ , 512 anchors, multi-scale density control. \* not significant at 0.05.

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## 227 Appendix B: sensitivity of the mismatch and sphere columns

228 Cross-fitting: each 2,048-patch is split at random into halves;  $\text{Hess}_M y$  is fitted on one half and local  
229  $R^2$  scored on the other (both directions shown), so the Hessian estimate and the outcome share no  
230 rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak  
231 ridge: the probe refit at  $\alpha = 1$  instead of 100 (global out-of-sample  $R^2$  rises by 0.10 to 0.15), which  
232 substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that  
233 probe.

## 234 Appendix C: relative second fundamental form, pilot

235 Two sphere-projected decoders (HSC =  $F$ , Legacy =  $G$ ; Phase 7 protocol), a global ridge map  $A$   
236 from  $x_F$  to  $x_G$  (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout  $R^2$  of  $A$ ;  
237 median first-order obstruction  $\|AJ_F - J_GL\|/\|AJ_F\|$  with  $L = J_G^+ AJ_F$ ; then multi-scale density-  
238 controlled partials against MKNN ( $k = 20$ ; log radius at  $k \in \{10, 30, 100, 300\}$  in both spaces)  
239 of each decoder’s  $\|H^S\|$ , the first-order obstruction,  $\|\Pi_G(L, L\cdot) - P_N^G A \Pi_F\|$  (because first-order  
240 matching  $AJ_F = J_G$  fails at the 44% level, the second derivative of the alignment residual is not  
241 coordinate invariant; in the decoder chart its target-normal quadratic coefficient additionally contains  
242  $-P_N^G AJ_F \Gamma^F$ , so relative II is reported only as an extrinsic diagnostic), the same with  $L$  fitted on 256  
243 neighbours’ latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment  
244 residual on  $F$ ’s tangent coordinates.

| label           | $d$ | mismatch, cross-fit |               | alignment, cross-fit |               | Hess. split cos | sphere, $\alpha = 1$    |
|-----------------|-----|---------------------|---------------|----------------------|---------------|-----------------|-------------------------|
|                 |     | fit A/score B       | fit B/score A | fit A/score B        | fit B/score A | p50             | (main: $\alpha = 100$ ) |
| mag_r           | 16  | -0.38               | -0.34         | +0.33                | +0.31         | +0.35           | -0.21 (-0.26)           |
| mag_r           | 20  | -0.43               | -0.40         | +0.39                | +0.37         | +0.24           | -0.22 (-0.26)           |
| photo_z         | 16  | -0.42               | -0.36         | +0.23                | +0.20         | +0.32           | +0.05* (-0.33)          |
| photo_z         | 20  | -0.42               | -0.42         | +0.22                | +0.22         | +0.20           | +0.03* (-0.36)          |
| smooth_fraction | 16  | -0.10               | -0.17         | +0.17                | +0.19         | +0.33           | -0.12 (-0.13)           |
| smooth_fraction | 20  | -0.15               | -0.21         | +0.05*               | +0.03*        | +0.23           | -0.13 (-0.15)           |
| stellar_mass    | 16  | +0.01*              | -0.04*        | -0.03*               | +0.06*        | +0.31           | -0.02* (-0.35)          |
| stellar_mass    | 20  | -0.06*              | -0.07*        | +0.01*               | -0.00*        | +0.19           | -0.02* (-0.35)          |

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

| run               | $R_A^2$ | tan. resid | $\ H^S\ $ HSC | $\ H^S\ $ Legacy | tan. resid | $\Pi_{\text{rel}}$ | $\Pi_{\text{rel}}$ (local $L$ ) | $\Pi_{\text{rel}}$ (data) |
|-------------------|---------|------------|---------------|------------------|------------|--------------------|---------------------------------|---------------------------|
| $d = 20$ , seed 0 | 0.64    | 0.44       | -0.15         | +0.04*           | -0.18      | +0.04*             | -0.18                           | +0.13                     |
| $d = 25$ , seed 0 | 0.64    | 0.43       | -0.10         | +0.06            | -0.16      | +0.05              | -0.23                           | +0.13                     |
| $d = 20$ , seed 1 | 0.64    | 0.44       | -0.11         | +0.05            | -0.19      | +0.04              | -0.20                           | +0.14                     |

Table 4: Relative-II pilot. \* not significant at 0.05.

## 245 Appendix D: the same test on four further encoders

246 Table 5 repeats Table 1 at  $d = 16$  on the same 86,471 galaxies embedded by four further encoders  
247 from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024,  
248 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed  
249 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors,  $k = 2,048$ ,  
250 multi-scale density control. The first row per label is the probe’s global out-of-sample  $R^2$ ; Table 6  
251 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per  
encoder.

| label           | quantity     | ViT-B  | DINOv3 | CLIP-B | ConvNeXt-B | ViT-L  |
|-----------------|--------------|--------|--------|--------|------------|--------|
| mag_r           | global $R^2$ | 0.52   | 0.57   | 0.50   | 0.52       | 0.52   |
|                 | shape        | +0.12  | -0.11  | +0.16  | +0.16      | -0.21  |
|                 | sphere       | -0.26  | -0.23  | -0.40  | -0.24      | -0.08* |
|                 | mismatch     | -0.39  | -0.57  | -0.34  | -0.32      | -0.44  |
|                 | alignment    | +0.35  | +0.53  | +0.32  | +0.21      | +0.13  |
| photo_z         | global $R^2$ | 0.51   | 0.56   | 0.49   | 0.49       | 0.51   |
|                 | shape        | -0.06* | -0.05* | -0.17  | +0.29      | -0.16  |
|                 | sphere       | -0.33  | -0.21  | -0.29  | -0.46      | -0.44  |
|                 | mismatch     | -0.45  | -0.61  | -0.45  | -0.48      | -0.46  |
|                 | alignment    | +0.26  | +0.33  | +0.06* | +0.22      | +0.14  |
| smooth_fraction | global $R^2$ | 0.53   | 0.59   | 0.49   | 0.56       | 0.55   |
|                 | shape        | +0.07* | -0.08* | +0.20  | +0.24      | +0.06* |
|                 | sphere       | -0.13  | -0.06* | +0.05* | -0.15      | -0.11  |
|                 | mismatch     | -0.09* | -0.33  | -0.25  | -0.20      | -0.01* |
|                 | alignment    | +0.19  | -0.02* | +0.24  | +0.04*     | +0.14  |
| stellar_mass    | global $R^2$ | 0.48   | 0.54   | 0.50   | 0.49       | 0.49   |
|                 | shape        | +0.00* | -0.15  | -0.10  | -0.11      | -0.13  |
|                 | sphere       | -0.35  | -0.26  | -0.38  | -0.40      | -0.38  |
|                 | mismatch     | -0.04* | +0.03* | -0.12  | -0.06*     | -0.02* |
|                 | alignment    | +0.01* | -0.05* | -0.01* | +0.07*     | -0.03* |

Table 5: Cross-encoder replication,  $d = 16$ , multi-scale density control. \* not significant at 0.05.

## 253 Appendix E: intervening on the readout’s normal component

254 The partials of Table 1 compare anchors with one another and cannot say whether reversing the probe-  
255 facing shape term hurts relative to removing it, since no anchor is flat. Here we score a counterfactual

| encoder    | label           | mismatch, cross-fit |               | alignment, cross-fit |               | Hess. split cos<br>p50 |
|------------|-----------------|---------------------|---------------|----------------------|---------------|------------------------|
|            |                 | fit A/score B       | fit B/score A | fit A/score B        | fit B/score A |                        |
| DINOv3     | mag_r           | −0.60               | −0.62         | +0.50                | +0.50         | +0.35                  |
|            | photo_z         | −0.56               | −0.59         | +0.28                | +0.27         | +0.24                  |
|            | smooth_fraction | −0.32               | −0.29         | −0.03*               | −0.02*        | +0.31                  |
|            | stellar_mass    | +0.03*              | +0.02*        | −0.03*               | −0.07*        | +0.22                  |
| CLIP-B     | mag_r           | −0.28               | −0.35         | +0.28                | +0.29         | +0.31                  |
|            | photo_z         | −0.41               | −0.40         | +0.08*               | +0.06*        | +0.25                  |
|            | smooth_fraction | −0.22               | −0.24         | +0.15                | +0.20         | +0.28                  |
|            | stellar_mass    | −0.14               | −0.13         | −0.04*               | +0.01*        | +0.19                  |
| ConvNeXt-B | mag_r           | −0.36               | −0.34         | +0.19                | +0.21         | +0.29                  |
|            | photo_z         | −0.46               | −0.45         | +0.18                | +0.19         | +0.30                  |
|            | smooth_fraction | −0.15               | −0.19         | −0.02*               | +0.03*        | +0.32                  |
|            | stellar_mass    | −0.05*              | −0.01*        | +0.14                | −0.03*        | +0.29                  |
| ViT-L      | mag_r           | −0.46               | −0.49         | +0.09*               | +0.15         | +0.30                  |
|            | photo_z         | −0.44               | −0.42         | +0.09*               | +0.13         | +0.24                  |
|            | smooth_fraction | −0.03*              | −0.04*        | +0.11                | +0.07*        | +0.30                  |
|            | stellar_mass    | −0.06*              | +0.01*        | −0.04*               | −0.03*        | +0.21                  |

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.

256 local second-order surrogate of the readout at a fixed manifold. At each anchor the fitted ridge weight  
 257 is split into tangent, radial and in-sphere normal parts  $w = w_T + w_{\text{rad}} + w_S$ ; we score the anchor's  
 258 2,048 neighbours by the data-side tangent-plus-sphere readout  $(w_T + w_{\text{rad}}) \cdot x$  plus  $tq$  with the  
 259 local intercept refit, where  $q = \frac{1}{2} \langle w_S, \Pi^S \rangle(u, u)$  is the decoder's in-sphere second-order term in the  
 260 anchor's chart coordinates:  $t = 1$  is the manifold's bending as seen in the readout's normal direction,  
 261  $t = -1$  its sign reversal,  $t = 0$  shape-flat (the sphere term  $-(w \cdot \hat{x})g$  is kept in the base). Because  
 262 the intercept is refit for every  $t$ , the local sum of squares is exactly  $\text{SSE}(t) = \|r_0 - tq_c\|^2$  with  $r_0$   
 263 the centred residual of the shape-flat predictor and  $q_c = q - \bar{q}\mathbf{1}$  the centred quadratic on the anchor's  
 264 neighbours; hence  $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$  and  $t = 1$  beats shape-flat iff  $2\langle r_0, q_c \rangle > \|q_c\|^2$ . This is  
 265 the finite-sample counterpart of the tensor-level condition  $2\langle R, K_S \rangle_g > \|K_S\|_g^2$  of Section 5, not  
 266 the same quantity: the tensor version needs the estimated label Hessian, the empirical one does not.  
 267 Columns: fraction of anchors where  $t = 1$  beats shape-flat (help) and where  $t = -1$  is worse than  
 268 shape-flat (hurt), the median change in local  $R^2$  at  $t = \pm 1$ , the median  $t^*$ ; then the same for a random  
 269 in-sphere normal direction  $v$  whose quadratic is rescaled to the same centred amplitude on the actual  
 270 neighbours,  $\|q_{v,c}\| = \|q_c\|$  (matching  $\|v\|$  to  $\|w_S\|$  instead leaves the contracted tensor at a fraction  
 271 of the fitted one, since  $\Pi^S$  spans at most  $d(d+1)/2$  of the  $\sim 750$  normal directions; matching  
 272  $\|\langle v, \Pi^S \rangle\|_g$  gives the same picture; Supplement 12). The anchors' neighbourhoods overlap (each  
 273 point lies in about twelve of them), so the fractions are not counts of independent trials; a maximal  
 274 set of anchors whose pairwise overlap is at most 5% of  $k$  has 19–21 members per run, on which  $t = 1$   
 275 beats shape-flat at 81–100% (one-sided sign test  $p \leq 4 \times 10^{-3}$  in every cell) and  $t = -1$  is worse  
 276 at 95–100% ( $p \leq 3 \times 10^{-5}$ ).

| run                  | label           | decoder in-sphere shape term |      |                  |                  |       | random orientation, matched $\ q_c\ $ |      |                  |                  |  |
|----------------------|-----------------|------------------------------|------|------------------|------------------|-------|---------------------------------------|------|------------------|------------------|--|
|                      |                 | help                         | hurt | $\Delta R^2(+1)$ | $\Delta R^2(-1)$ | $t^*$ | help                                  | hurt | $\Delta R^2(+1)$ | $\Delta R^2(-1)$ |  |
| ViT-B, $d = 16$      | mag_r           | 0.97                         | 1.00 | +0.030           | -0.051           | 2.3   | 0.26                                  | 0.80 | -0.009           | -0.009           |  |
|                      | photo_z         | 0.99                         | 0.99 | +0.028           | -0.038           | 3.0   | 0.33                                  | 0.68 | -0.005           | -0.006           |  |
|                      | smooth_fraction | 0.90                         | 1.00 | +0.021           | -0.040           | 1.7   | 0.24                                  | 0.78 | -0.008           | -0.010           |  |
|                      | stellar_mass    | 0.96                         | 0.99 | +0.026           | -0.040           | 2.6   | 0.29                                  | 0.70 | -0.006           | -0.006           |  |
| ViT-B, $d = 20$      | mag_r           | 0.99                         | 1.00 | +0.031           | -0.049           | 2.6   | 0.24                                  | 0.78 | -0.007           | -0.008           |  |
|                      | photo_z         | 0.99                         | 1.00 | +0.029           | -0.038           | 3.6   | 0.36                                  | 0.71 | -0.004           | -0.005           |  |
|                      | smooth_fraction | 0.90                         | 1.00 | +0.020           | -0.035           | 1.9   | 0.22                                  | 0.77 | -0.007           | -0.008           |  |
|                      | stellar_mass    | 0.97                         | 1.00 | +0.027           | -0.038           | 3.1   | 0.32                                  | 0.72 | -0.005           | -0.005           |  |
| DINOv3, $d = 16$     | mag_r           | 0.92                         | 0.99 | +0.018           | -0.031           | 2.0   | 0.29                                  | 0.71 | -0.007           | -0.006           |  |
|                      | photo_z         | 0.96                         | 0.99 | +0.020           | -0.030           | 2.6   | 0.29                                  | 0.71 | -0.006           | -0.004           |  |
|                      | smooth_fraction | 0.93                         | 1.00 | +0.024           | -0.055           | 1.5   | 0.22                                  | 0.78 | -0.014           | -0.013           |  |
|                      | stellar_mass    | 0.95                         | 1.00 | +0.016           | -0.028           | 2.1   | 0.26                                  | 0.74 | -0.005           | -0.005           |  |
| CLIP-B, $d = 16$     | mag_r           | 0.95                         | 0.99 | +0.018           | -0.031           | 2.2   | 0.30                                  | 0.68 | -0.006           | -0.005           |  |
|                      | photo_z         | 0.99                         | 1.00 | +0.019           | -0.026           | 3.4   | 0.36                                  | 0.64 | -0.003           | -0.003           |  |
|                      | smooth_fraction | 0.92                         | 0.99 | +0.014           | -0.027           | 1.7   | 0.25                                  | 0.69 | -0.006           | -0.005           |  |
|                      | stellar_mass    | 0.97                         | 1.00 | +0.020           | -0.028           | 2.8   | 0.35                                  | 0.68 | -0.004           | -0.005           |  |
| ConvNeXt-B, $d = 16$ | mag_r           | 0.92                         | 1.00 | +0.018           | -0.031           | 2.0   | 0.26                                  | 0.73 | -0.007           | -0.007           |  |
|                      | photo_z         | 0.99                         | 1.00 | +0.018           | -0.026           | 2.9   | 0.28                                  | 0.64 | -0.004           | -0.003           |  |
|                      | smooth_fraction | 0.88                         | 0.98 | +0.016           | -0.036           | 1.6   | 0.23                                  | 0.75 | -0.009           | -0.008           |  |
|                      | stellar_mass    | 0.99                         | 1.00 | +0.022           | -0.032           | 3.1   | 0.29                                  | 0.65 | -0.005           | -0.003           |  |
| ViT-L, $d = 16$      | mag_r           | 0.96                         | 0.99 | +0.022           | -0.034           | 2.6   | 0.32                                  | 0.74 | -0.005           | -0.006           |  |
|                      | photo_z         | 0.97                         | 0.99 | +0.018           | -0.025           | 3.2   | 0.34                                  | 0.67 | -0.004           | -0.003           |  |
|                      | smooth_fraction | 0.79                         | 0.97 | +0.013           | -0.029           | 1.6   | 0.25                                  | 0.77 | -0.006           | -0.008           |  |
|                      | stellar_mass    | 0.98                         | 1.00 | +0.022           | -0.032           | 2.9   | 0.30                                  | 0.67 | -0.004           | -0.004           |  |

Table 7: Counterfactual scaling of the in-sphere shape term in a local second-order surrogate of the readout, 512 anchors per run. The manifold’s bending as seen in the readout’s normal direction helps at a large majority of anchors, its sign reversal hurts at nearly every anchor, a random orientation of the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign;  $t^* > 1$  throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial term.